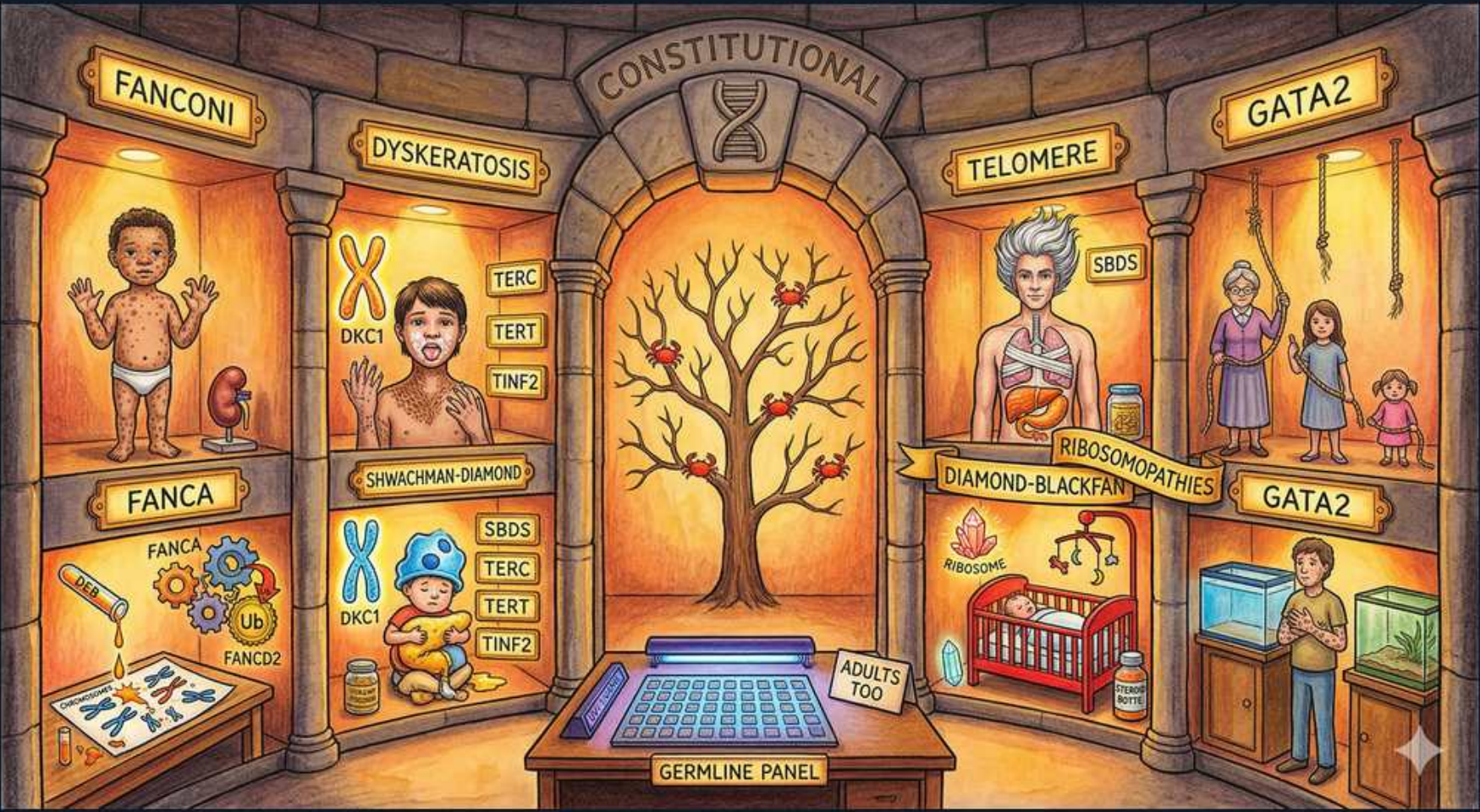


Bone Marrow Failure — The Constitutional Archives (Inherited BMF)



1. Constitutional archway

WHAT YOU SEE

Banner reading CONSTITUTIONAL with a DNA helix

WHAT IT ENCODES

Inherited (constitutional) bone marrow failure syndromes are germline-driven defects in DNA repair, telomere maintenance, or ribosome biogenesis that classically present in childhood with cytopenias plus characteristic physical anomalies. Unlike acquired aplastic anemia, they carry a lifelong predisposition to MDS/AML and solid tumors. Suspect them in any young patient with cytopenias plus dysmorphic features, or in an adult with unexplained marrow failure and a suggestive family history.

BOARD TRAP

Do NOT default to acquired aplastic anemia in a child or young adult with cytopenias — a constitutional syndrome must be excluded first, because it changes cancer surveillance, transplant conditioning (these patients are exquisitely sensitive to chemo/radiation), and donor selection (an affected sibling cannot donate).

2. Fanconi child

WHAT YOU SEE

FANCONI placard above a short-statured child with skin findings

WHAT IT ENCODES

Fanconi anemia is the most common inherited BMF syndrome — autosomal recessive (rarely X-linked FANCB), with mutations in FANC genes impairing the DNA interstrand crosslink repair pathway. Phenotype mnemonic: short stature, café-au-lait macules (hyperpigmentation), thumb/radial ray anomalies, microcephaly, and hypogonadism. Progressive marrow failure typically emerges by age 5-10, and the lifetime risk of AML and squamous cell carcinomas (head/neck, gynecologic) is markedly elevated.

BOARD TRAP

Up to 30% of Fanconi patients have NO congenital anomalies — a normal-looking child with aplastic anemia still needs a chromosome breakage test before being labeled idiopathic aplastic anemia.

3. Thumb / radial anomaly

WHAT YOU SEE

Child's malformed hand and small kidney icon

WHAT IT ENCODES

Absent or hypoplastic thumbs and radii are the classic skeletal signature of Fanconi anemia (the thumb is involved, distinguishing it from TAR syndrome where thumbs are spared). Renal anomalies — horseshoe kidney, ectopic or absent kidney — accompany the radial-ray defect (think VACTERL-like overlap).

BOARD TRAP

Thumb anomaly + thrombocytopenia is NOT automatically Fanconi: in TAR syndrome (thrombocytopenia-absent radius) the thumbs are PRESENT, whereas Fanconi involves the thumb itself. Use thumb status to separate them.

4. FANCA / FANCD2

WHAT YOU SEE

FANCA gears with Ub and FANCD2 ubiquitin tag

WHAT IT ENCODES

The FA core complex (FANCA is the most common subtype, ~60-70% of cases) mono-ubiquitinates FANCD2/FANCI, which then recruits downstream repair proteins (including BRCA2/FANCD1) to resolve DNA interstrand crosslinks at stalled replication forks. Loss of this pathway leaves cells unable to repair crosslinks, producing genomic instability and cancer predisposition.

BOARD TRAP

FANCD1 is BRCA2 — biallelic BRCA2 mutations cause a severe Fanconi phenotype with early childhood solid tumors (medulloblastoma, Wilms). Don't think of BRCA2 only as adult breast/ovarian cancer.

5. Chromosome fragility

WHAT YOU SEE

Broken/fragmented chromosomes on a tray

WHAT IT ENCODES

The diagnostic test for Fanconi anemia is the chromosome breakage / fragility test: patient lymphocytes are exposed to DNA crosslinking agents — diepoxybutane (DEB) or mitomycin C (MMC) — and show dramatically increased chromosomal breaks and radial figures versus controls. This is the gold-standard screen before confirmatory germline gene panel.

BOARD TRAP

The provocative agents are DEB and mitomycin C specifically — not generic radiation. A normal spontaneous karyotype does NOT exclude Fanconi; breakage only appears after crosslinker challenge.

6. Dyskeratosis triad

WHAT YOU SEE

DYSKERATOSIS sign above a child with mucosal/skin findings

WHAT IT ENCODES

Dyskeratosis congenita classic mucocutaneous triad: (1) abnormal skin pigmentation (reticular/lacy hyperpigmentation of neck and upper chest), (2) nail dystrophy, (3) oral leukoplakia. Marrow failure is the leading cause of death, but patients also develop pulmonary fibrosis, hepatic fibrosis/cirrhosis, and a high risk of MDS/AML and squamous cell carcinoma (oral leukoplakia is premalignant).

BOARD TRAP

The leukoplakia is premalignant — don't dismiss oral white patches as candidiasis. And don't confuse DC's reticular pigmentation with the discrete café-au-lait spots of Fanconi.

7. DKC1 telomere genes

WHAT YOU SEE

X-chromosome with DKC1, plus TERC/TERT/TINF2 tags

WHAT IT ENCODES

Dyskeratosis congenita is a telomere maintenance disorder. The most severe X-linked form is caused by DKC1 (encodes dyskerin, which stabilizes the telomerase RNA component); autosomal forms involve TERC (telomerase RNA), TERT (telomerase reverse transcriptase), and TINF2 (shelterin). The result is critically short telomeres, measurable by flow-FISH of leukocytes (very short telomeres for age is the screening test).

BOARD TRAP

Telomere disease shows genetic anticipation — successive generations present earlier and more severely because shortened telomeres are inherited. A mild adult parent can have a severely affected child.

8. Telomere bench

WHAT YOU SEE

TELOMERE sign over a prematurely grey-haired man

WHAT IT ENCODES

Short-telomere syndromes produce a premature-aging phenotype: early greying, marrow failure, and progressive fibrosis of lung (idiopathic pulmonary fibrosis) and liver. In adults, a telomere defect (TERT/TERC) may present as familial pulmonary fibrosis or unexplained cirrhosis with subtle cytopenias and macrocytosis — the marrow failure can be mild or late.

BOARD TRAP

Telomere biology disorders are a recognized cause of adult-onset idiopathic pulmonary fibrosis and cryptogenic cirrhosis — these patients tolerate reduced-intensity transplant conditioning poorly and are prone to fatal pulmonary complications.

9. Adults too tree

WHAT YOU SEE

Central archive tree with red apples and ADULTS TOO label

WHAT IT ENCODES

Constitutional BMF — especially telomere disorders and some Fanconi/GATA2 cases — is NOT exclusively pediatric. Adults can present de novo with aplastic anemia, MDS, pulmonary/hepatic fibrosis, or unexplained macrocytic cytopenias, and germline testing in adults with 'idiopathic' marrow failure increasingly uncovers these syndromes.

BOARD TRAP

Assuming inherited BMF 'only presents in children' is a classic exam trap — missing an adult telomere or GATA2 syndrome leads to wrong transplant conditioning and use of an affected related donor.

10. Diamond-Blackfan crib

WHAT YOU SEE

DIAMOND-BLACKFAN sign with red baby in a crib, RIBOSOME

WHAT IT ENCODES

Diamond-Blackfan anemia is a congenital PURE RED CELL aplasia (a ribosomopathy) presenting in the first year of life with macrocytic anemia, reticulocytopenia, and absent marrow erythroid precursors. Caused by ribosomal protein gene mutations (RPS19 most common). Labs: elevated HbF, elevated erythrocyte adenosine deaminase (eADA). Anomalies include craniofacial defects and the triphalangeal thumb. First-line treatment is corticosteroids; transfusion-dependent or steroid-refractory cases go to transplant.

BOARD TRAP

Diamond-Blackfan is RED-CELL-ONLY (anemia with normal WBC/platelets) — it does NOT cause pancytopenia. Distinguish from transient erythroblastopenia of childhood (TEC): DBA presents <1yr with macrocytosis, ↑HbF, ↑eADA and congenital anomalies, whereas TEC occurs in older toddlers, is normocytic, and self-resolves.

11. Shwachman-Diamond

WHAT YOU SEE

SHWACHMAN-DIAMOND label with SBDS toddler

WHAT IT ENCODES

Shwachman-Diamond syndrome: autosomal recessive SBDS gene mutation (a ribosome-assembly defect) producing the triad of neutropenia (the dominant cytopenia, with recurrent infections), exocrine pancreatic insufficiency (steatorrhea, failure to thrive, low fecal elastase), and skeletal abnormalities (metaphyseal dysostosis). Carries a significant risk of progression to MDS/AML.

BOARD TRAP

The marrow problem here is primarily NEUTROPENIA, not anemia — and the pancreatic insufficiency can mimic cystic fibrosis, but sweat chloride is normal and there is neutropenia. Don't confuse with cystic fibrosis.

12. GATA2 family

WHAT YOU SEE

GATA2 sign with a family group

WHAT IT ENCODES

GATA2 deficiency is an autosomal dominant transcription-factor disorder causing inherited predisposition to MDS/AML, profound monocytopenia, B/NK lymphopenia, and severe susceptibility to disseminated mycobacterial, fungal, and HPV infections (the 'MonoMAC' syndrome). Pulmonary alveolar proteinosis and lymphedema (Emberger syndrome) are associated. Often presents in adolescence/young adulthood.

BOARD TRAP

Persistent monocytopenia with atypical/disseminated infections (NTM, HPV warts) plus a family history of MDS = think GATA2 — it is a frequently overlooked adult/adolescent cause of familial MDS and is curable only by transplant.

13. Germline panel

WHAT YOU SEE

Glowing GERMLINE PANEL console at the base

WHAT IT ENCODES

Workup of suspected constitutional BMF combines targeted functional screens (chromosome breakage for Fanconi, leukocyte telomere length by flow-FISH for dyskeratosis congenita) with a comprehensive germline next-generation sequencing panel that pins the exact syndrome and gene. This drives cancer surveillance, transplant conditioning intensity, and — critically — exclusion of affected relatives as stem-cell donors.

BOARD TRAP

Always send germline (not just tumor/marrow) testing and screen potential RELATED donors — an unaffected-appearing sibling may carry the same germline mutation, making them an unsafe donor and risking donor-derived disease.